For the computation of confidence intervals and for testing of significance, the critical values Z_α at the different level of significance is given in the following table:

Normal Probability Table

Critical Values Z_α	Level of significance (α)			
	1%	2%	5%	10%
Two-tailed test	$\lvert Z_\alpha \rvert$=2.58	$\lvert Z_\alpha \rvert$=2.33	$\lvert Z_\alpha \rvert$=1.96	$\lvert Z_\alpha \rvert$=1.645
Right tailed test	Z_α =2.33	Z_α =2.055	Z_α =1.645	Z_α =1.28
Left tailed test	Z_α = -2.33	Z_α = –2.055	Z_α = –1.645	Z_α = –1.28

The calculation of confidence interval is illustrated below.

Example 8.11

A machine produces a component of a product with a standard deviation of 1.6 cm in length. A random sample of 64 componentsvwas selected from the output and this sample has a mean length of 90 cm. The customer will reject the part if it is either less than 88 cm or more than 92 cm. Does the 95% confidence interval for the true mean length of all the components produced ensure acceptance by the customer?

Solution:

Here φ is the mean length of the components in the population.

The formula for the confidence interval is

$$\bar{x} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

Here σ = 1.6, $Z_{\alpha/2}$ = 1.96, $\bar{x}$ = 90 and n = 64

Then $$S.E. = \frac{\sigma}{\sqrt{n}} = \frac{1.6}{\sqrt{64}} = 0.2$$

Therefore, $90 - (1.96 \times 0.2) \leq \varphi \leq 90 + (1.96 \times 0.2)$

i.e. $(89.61 \leq \varphi \leq 90.39)$

This implies that the probability that the true value of the population mean length of the components will fall in this interval (89.61,90.39) at 95% . Hence we concluded that 95% confidence interval ensures acceptance of the component by the consumer.

Example 8.12

A sample of 100 measurements at breaking strength of cotton thread gave a mean of 7.4 and a standard deviation of 1.2 gms. Find 95% confidence limits for the mean breaking strength of cotton thread.

Solution:

Given, sample size = 100, $\bar{x}$ = 7.4, since σ is unknown but s = 1.2 is known.

In this problem, we consider $\breve{\sigma} = s$, $Z_{\alpha/2} = 1.96$

$$S.E. = \frac{\breve{\sigma}}{\sqrt{n}} = \frac{s}{\sqrt{n}} = \frac{1.2}{\sqrt{100}} = 0.12$$

Hence 95% confidence limits for the population mean are

$$\bar{x} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

$$7.4 - (1.96 \times 0.12) \leq \mu \leq 7.4 + (1.96 \times 0.12)$$

$$7.4 - 0.2352 \leq \mu \leq 7.4 + 0.2352$$

$$7.165 \leq \mu \leq 7.635$$

This implies that the probability that the true value of the population mean breaking strength of the cotton threads will fall in this interval (7.165,7.635) at 95% .

Example 8.13

The mean life time of a sample of 169 light bulbs manufactured by a company is found to

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